CS24301 — THEORY (3.0 CR) | 10-PAGE MASTER REVISION | BIT MESRA | NEP SCHEME

Compiler Design — 10-Page Master Quick Revision

High-Yield Formula Sheet, Decision Trees & Top 10 BIT Mesra PYQ Solutions

🏛️ 10-Page Master Quick Revision — Compiler Design (CS24301)

⚡ High-Yield Formula & Rule Master Matrix

Core Concept	Crucial Formula / Rule	Exam Trap / Memory Key
Direct DFA: Followpos	Cat-node $c_1 \cdot c_2$: $\text{followpos}(i) \cup= \text{firstpos}(c_2)$ for $i \in \text{lastpos}(c_1)$. Star-node c_1^*: $\text{followpos}(i) \cup= \text{firstpos}(c_1)$ for $i \in \text{lastpos}(c_1)$.	Must augment $(r)\#$; accepting states contain $\#$.
LL(1) Table Rule	For $A \to \alpha$: add to $M[A, a]$ for $a \in \text{FIRST}(\alpha)$. If $\epsilon \in \text{FIRST}(\alpha)$, add to $M[A, b]$ for $b \in \text{FOLLOW}(A)$.	No duplicate entries permitted in any cell.
LR Parser Hierarchy	$\text{LL}(1) < \text{SLR}(1) < \text{LALR}(1) < \text{CLR}(1)$	$\text{LALR}(1)$ merges core states of $\text{CLR}(1)$, reducing tables to $\text{SLR}(1)$ size.
2D Array Row-Major	$\text{Base} + ((i - low_1) \times n_2 + (j - low_2)) \times w$	Multiply by n_2 (number of columns) for row-major.
2D Array Col-Major	$\text{Base} + ((j - low_2) \times n_1 + (i - low_1)) \times w$	Multiply by n_1 (number of rows) for col-major.
Cyclomatic Complexity	$V(G) = E - N + 2 = P + 1 = \text{Regions}$	Count decision/predicate nodes P.

🔥 Top 10 High-Probability BIT Mesra Exam Questions

Q1. Explain the 6 phases of compilation with the translation of position = initial + rate * 60.

1. Lexer: Token stream with symbol table pointers. 2. Parser: Hierarchical syntax tree. 3. Semantic: Type checks and inttofloat coercion. 4. ICG: Three-address code with temporaries. 5. Optimizer: Constant folding (60.0) and temporary reduction. 6. Code Gen: Target register assembly instructions (LDF, MULF, ADDF, STF).

Q2. State the difference between S-Attributed and L-Attributed SDDs.

S-Attributed: Exclusively synthesized attributes; evaluated bottom-up during post-order traversal.
L-Attributed: Inherited attributes allowed only if they depend on parent or left-siblings; evaluated in a single top-down left-to-right DFS pass.

Q3. Detail the contents of an Activation Record and explain static vs. dynamic links.

Contents: Parameters, Return Value, Return Address (PC), Dynamic Link (caller AR pointer), Static Link (lexical parent AR pointer), Saved Machine Registers, Local Variables, Temporaries.